

POSTURE

→ "Attitude assumed by body as a result of co-ordinated action performed by a group of muscles acting to maintain stability" - S Posture.

Static - Body / segments don't move
like - Standing, Sitting

Dynamic - Body / segments move
like - Running, Walking

Postural Control -

→ Here muscle plays a major role.

→ Ability to maintain stability of body / segments in response to forces that threaten the Equilibrium.

External force and Internal force -

External force - Reactive response occurs as reaction

(or)
Reactive or Compensatory.

Internal force - Proactive response occurs in anticipation
(or)
Anticipatory.

Muscle - Strength / ROM -

→ CNS - Interprets, Organises inputs from various region and selects response.

1° - Proprioception - Hip / Trunk.

- 2° - Vestibular System
3° - Vision.

Goals for Postural Control -

- Controls body orientation in space
- Maintains center of mass over base of support
- Stabilize Head - Eyes at the same level through Vision and Vestibular System.

Muscle Synergy -

- Perturbation is any sudden change in conditions that displaces the body posture away from equilibrium.
- Perturbation $\begin{cases} \rightarrow \text{Sensory Perturbation} \\ \rightarrow \text{Mechanical Perturbation} \end{cases}$

Sensory Perturbation - It might happen by altering visual input.

Mechanical Perturbation - These are displacements that involve direct ~~but~~ changes in the relationship of body's COM to the BOS.

- Postural responses to perturbations caused by either platform movement or by pushes and pulls are reactive or compensatory responses in that they are involuntary reactions.

Fixed-Support Synergies -

- Pattern of muscle activity in which base of support

is fixed during perturbation and recovery of equilibrium.

→ Two types of Synergy -

1) Ankle Synergy (distal - proximal)

2) Hip Synergy (proximal - distal)

→ Fixed-Support Synergy is used mostly when it is not possible to use change in support strategy.

Ankle Synergy

Backward motion of Body -

→ Plantar flexion of foot

^(Tibialis anterior) Dorsiflexion muscles get fired to get ankle ft to neutral position.

→ Knee flexion

For Extension all quadrup muscles get fired to get knee into extension

→ Hip Extension.

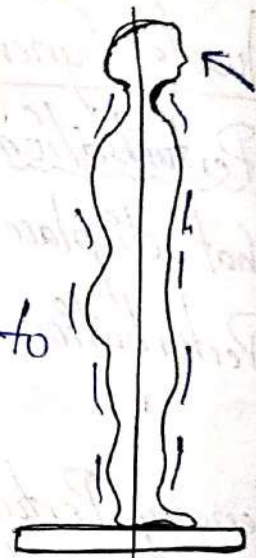
Rectus femoris and abdominis get fired to flexion near Hip.

→ Cervical Neck help to get back to neutral position.

Forward motion of body -

→ Dorsiflexion of foot

Plantar flexion muscles get fired to get ankle ft



to neutral position (Calf muscles).

→ Knee Extension

For flexion all Hamstrings muscles get fired to get knee into flexion.

→ Hip flexion, Hamstrings muscles get fired to get Hip flexion. (Glutes)

→ Paraspinal muscles and Neck Extension.

Hip Synergy

Forward translation of Support -

Abdominals

Quadriceps, Rectus femoris

Tibialis Anterior

Backward translation of Support -

Paraspinals

Hamstrings

Gastrocnemius

** When base is moved or pulled forward the object falls backward that is forward translation of support.
and base is pulled backward the object falls forward that is backward translation of support.

Change in Support Strategies

Stepping / Grasping - ↑ base of support so that COM remains within BOS.

→ Only synergies that are successful in maintaining equilibrium.

→ Young subjects take single step while elderly take multiple steps which are shorter and less in height.

→ Elderly have difficulty in maintaining lateral stability, when stepping foot is off the ground →

Hip Abductor strength (weak)

→ Elderly Rely more on arm reaction than young, But are slow at it.

Head Stabilizing Strategy

→ Proactive Strategy -

Because it occurs in anticipation of initial forces produced to change body position.

→ 2 types of Head Stabilizing.

① HSS - Head Stabilization in Space

② HST - Head Stabilization in Trunk.

→ Postural control is organised in more of reflex like manner with R primary supraspinal control.

→ (Spinal cord is not much involved in postural

control but more often supraspinal (CNS) control the body posture).

Kinetics -

Internal Forces -

They are produced by muscle activity, passive ligament, tendons, of capsule and soft tissue structure.

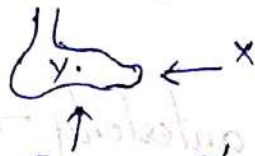
External Forces -

That consist of inertia, Gravity, Ground Reaction force.

Gravity - Downward pull of the earth on our body.

GRF - Ground pushes body up magnitude and direction.

GRF has three components x, y, z Axis



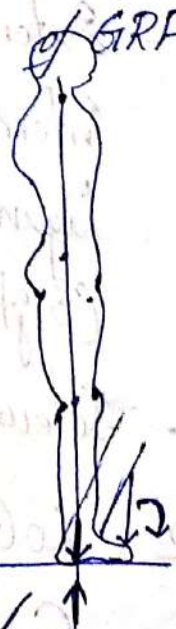
In standing its opposite to GRF.

Center of Pressure - Point of application of GRF

(Theoretical point).

- Movement in CoP can tell a lot about postural sway.

→ During standing a slight sway ahead and external gravitational torque are countered / counter balance by internal



Torque created by muscle at the jt.
Example: An external dorsiflexion moment is
countered / counter balance by internal plantar

flexion moment.
Optimum posture :-
Postural Analysis - Lateral View

Ankle - LoG goes Anteriorly to the axis.

External dorsiflexion torque
are countered by internal
plantar flexor torque.

↓
Soleus, Gastrocnemius
Control anterior sway.

- No ligament is counterbalance
force.



Knee - LoG goes anteriorly to the axis.

- External extension moment will be fighting with
internal flexion moment by posterior capsule,
ligament, Hamstring (flexion), Gastrocnemius
(2nd jt muscle (knee flexion)).

- Soleus will cause extension at knee.

Hip - LoG goes posteriorly to the axis. Through
Greater trochanter.

- External extension $\rightarrow$ (Posterior pelvic tilt)
- Internal Flexion $\rightarrow$ (Iliopsoas muscle)
- (ischiofemoral, iliofemoral, pubofemoral ligaments)
- $\rightarrow$ In optimal posture $\rightarrow$ pelvis is level (ASIS) $\neq$ (PSIS) - But Sway Back.

$\rightarrow$ LoG goes posteriorly $\uparrow\uparrow$, posterior tilt of pelvis - Hip extension ($\downarrow$ Glutes activation).

Flat lumbar spine, kyphosis (hump). Thoracic spine forward head posture.

$\rightarrow$ $\uparrow$ tension in anterior hip ligaments

$\rightarrow$ $\uparrow$ torque on all the jts.

Knee - Hyper extension

Hip - Extension

Flattening of lumbar spine

Kyphotic at thoracic spine

Forward head posture.

Lateral Analysis of Upper Body -

SI JT - LoG Anterior

- External moment - Nutation (Gap b/w Sacrum $\neq$ pelvic will be $\uparrow$ ing during Nutation).

- Internal moment - Counter Nutation.

- Active - (Muscles) Transverse abdominis

- Passive - (Ligaments) Sacrotuberous, Sacrospinous, Iliolumbar lig, Anterior SI lig.

Lumbar Spine - LOG Posterior

- External moment - Extension
- Flexion created by abdominals & Obliques.
- Passively resisted by anterior longitudinal ligament, ilio-lumbar ligament, Annulus fibrosus, & capsule prevents extension movement.

Thoracic Spine - LOG Anteriorly.

- External flexion moment resisted by posterior longitudinal ligament, Inter spinous ligament.
- Thoracic spinal extensors.

Cervical Spine - LOG Posteriorly.

- External extension torque countered by Neck flexors and ligaments like Anterior Scalene, Longus capitis, C6 anterior longitudinal ligament, Annulus fibrosus, Capsule.

Atlanto-Occipital Jt - LOG Anteriorly.

- External flexion moment countered by extensors - (Deep cervical muscles).
- Rectus capitis, Semispinalis - Capitis / cervicis
- Passive - Ligament Nuchae, Alar ligament, atlanto-occipital, Atlanto occipital membrane.

Postural Deviations - Lateral View

Hammer Toe deformative

Lower body

MTP - Hyperextension

PIP - Flexion

DIP - Extension



→ Callusitis on PIP (superiorly) 1st phalanx

Distal tip of phalanx

Flexors stretched at MTP jt & short at PIP jt.

Cause - Long / Short toe extensor / lumbricals paralysed.

Claw Toe deformative

MTP - Hyper extension

PIP & DIP - flexion



BOS ↓, ↑ postural sway, ↓ stability.

→ Callus formation under (MTP head) or (Dorsal region of distal phalanx).

→ Proximal phalanx can subluxate dorsally. In superior direction.

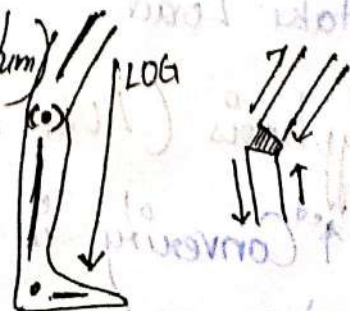
Cause - Restrictive footwear, Neuromuscular disorder or weak intrinsic foot muscles.

Line of Gravity is Anterior to the knee

→ Hyperextension knee (Genu recurvatum)

→ All the load, on posterior jt capsule + ligament.

→ Jt capsule adaptive lengthening.



↓ stability.

Occur due to ↓ dorsiflexion (plantar flexor contraction)

Flexed knee Posture (LOG posteriorly)

→ LOG posteriorly to knee - Quadriceps activity is needed to maintain position.

↑ compressive stress on knee +
fatigue = hip - flexion - ↑ extensor
activity = Ankle - Dorsiflexion.

↑ Soleus activity.



Lateral View - Upper body

Anterior Pelvic Tilt + Lordosis

↑ extensor activity - LOG anterior

Balanced by Kyphosis at Thorax.

Optimum Lordosis Helps in Nutrition
and Fluid movement at Disc.

Posterior Pelvic Tilt + ↓ Lordosis

↓ Flexibility + Reduced Ability to
take Load.

Kyphosis (LOG Anterior)

→ ↑ Convexity in posterior region

↑ Concavity in anterior region



- Thoracic, Sacral region
- Compression in anterior region
Distraction in posterior region
- Dowgers Hump } caused due to Osteoporosis
Gibbus deformity } and Tuberculosis.

Lordosis (L0G Posteriorly)

- ↑ ed concavity posterior region
↑ ed convexity anterior region
- Compression in posteriorly
Distraction in anteriorly.
- Cervical, lumbar region.

Forward Head Posture

- Head anterior + ↑ cervical lordosis + Compressⁿ
of posterior zygapophyseal jt + Disc. Compression +
↓ Space (intervertebral foramen).
- ↑ Extensor activity ← ischemic muscle due to
constant isometric contraction to counter external
flexion torque.
- Shoulder medially rotated + ↑ kyphosis, ↓ vital
capacity, ↓ height, Can effect TMJ too.

Anterior View

- Head - LOG passes through middle, eyes & ears are on the same level.
- Right and left angles b/w shoulder & neck should be symmetrical and relative external rotation.
- In Hips LOG passes through umbilicus & passes b/w the ASIS is at the same level LOG passes b/w the knee (equidistant) with patella falling front symmetrical LOG passes equidistant from medial malleoli.

Postural Posterior View

- LOG passes through middle of Head.
- Arms hang naturally at side with equal distance b/w arm and body on either side.
- Scapula lie flat on ribcage (equidistant)
- PSIS and gluteal fold symmetrical.
- Knees should be level, and equidistant.
- Heel and TA should be vertical and malleoli equidistant.

Antero-Posterior View (Lower Limb)

Ankle - Pes Planus (Flat Foot)

Calcaneal eversion, reduction or absent of medial arch.

→ 2 types of flat foot

1) Rigid

2) Flexible

→ It is a structural problem

→ It is a functional problem

→ Present at wt bearing & Non-wt bearing

→ Present at wt bearing

Pes Cavus
Calcaneal Inversion, Very rigid foot that is unable to adapt to uneven surfaces.

→ Foot stuck in supination. Hence bad at shock absorption.

*** Way to determine that we have correct foot or not?

→ By Hees line method.

Mark the medial malleolus & 1st MT prominence and join the line.

Then see the navicular bone where it falls if it is above the Hees line or below the Hees line.

Knee - Genu Valgum (Knock knees)

→ Seen in children of 2-6 yrs.

→ ↑ external torque on knee with tensile stress on medial structure and compressive forces on lateral structures.

→ Patella laterally displaced & subjected to dislocat?

Causes - Weak Abductors.

Genu Varum (Bow Legs)

- Seen in children 3 to 4 yrs.
- Cortical thickening and compression in medial compartment of the knee.

Causes - Vit. D deficiency, rickets, Osteochondritis (It is a painful type of osteochondrosis where the cartilage or bone in a jt is inflamed).

Antero-Posterior View (Upper Body)

Scoliosis

- Unequal shoulder level
 - Unequal waist angles
 - lateral spinal curvature not erect, normally it should co-inside with L6 Rib hump.
- Pay attention to body contours as early as 8 yrs. old child.

Home Screening every 6 months till age of 14 yrs.

To identify and recognise AIS and understand more about this disease

(Adolescent Idiopathic Scoliosis)

→ Scoliosis is divided into Structural & functional
→ Functional can be reversed if cause is fixed.

Ex: Leg length Discrepancy or muscle spasm.

→ Structural → Changes in bone, soft tissue.

→ Named after direction of convexity and the location of the curve.

= Right Thoracic left lumbar Scoliosis

→ They are 2 types of curve

① Double major Curve

Two structural curves of same size

② Tripple curve

Three region of a vertebral column

Adolescent Idiopathic Scoliosis

→ Mostly seen in mens.

→ Lateral curvature + Deformity in frontal + Sagi-Hal plane.

→ Age of Onset = Infantile - (0-3) yrs

Juvenile - (4-10) yrs

Adolescent - (>10) yrs.

→ Muscular / Ligamentous / Fascial support system defect.

External lateral flexion moment.

→ Asymmetrical loading of intervertebral disc.

→ Deformation of disc → High abnormal pressure

Wedging of disc → Deviation of vertebral with rotation.

- Growth inhibition of vertebra on Compression
- side → wedging of vertebra → end plate calcification
- Affects disc nutrition.
- Head out of line with sacrum → Compensatory Curve → Adaptive shortening of muscles → lengthening of ligaments capsule

Sitting

Goals :- Stable position.

↓ Energy expenditure.

Hard to standardize because of various postures.

Each posture will vary in Muscular activity with LoG.

Inter-Discal pressure.

Seat interface pressure.

Types of Sitting

Active sitting

Relaxed sitting

Slump sitting

Slouched sitting

Kinetics and Kinematics

In Slump and Slouched sitting -

LoG - Anteriorly → Flexion relaxed - no muscle

→ Activity - all the load on passive structure (capsule, ligaments).

→ Instead in active and relaxed sitting muscle activity increases.

→ Lordosis is reduced in all due to posterior pelvic tilt.

→ ↑ intra discal pressure → Anterior Annulus compression
→ Posterior annulus distraction

Disc Degeneration ← Stretches thin ← Sustained Load

Sitting v/s Standing -

→ Erect sitting with lordosis has higher extensor activity than standing → posterior pelvic tilt in sitting make muscles work harder.

higher Extensor activity → Can be reduced with arm support.

→ Intradiscal Pressure

→ ↑ pressure in erect sitting > Standing.

→ Co-contraction muscles ↑ Compression at (L₄-L₅)

But long hrs of standing ↓ disc height, in standing more than sitting.

→ Moving around reduces disc shrinkage.

Eg: Dynamic chair can be used.

Seat Interface

→ In sitting, force will be transmitted through

Hips, Ischial tuberosity.

→ If person is thin or ↓ muscle and fat in contact region ↓.

↑ pressure at Ischial region - Sores (not same with foot).

→ Pressure depends on - Foot rest (It ↓ pressure)
Foam cushion of 8cm

Time of exposure.

→ ↑ pressure can be countered by

8cm Cushion - foam

Change in position flexion, lateral flexion.

Push-ups - upper body

Strength (*Latissimus dorsi*)

Posture in Early Years

→ Infants develop the control and curve lordosis at cervical spine - Head control.

Lordosis at lumbar spine - sit to stand.

Co-contraction of agonist and antagonist.

Alignment changes (vertical) b/w age of 4 to 12 yrs.

Forward Head, Thoracic kyphosis ↑, lumbar

lordosis ↓.

Posture in Elderly

→ Posture goes for overall flexion.

Cervical lordosis ↑, ↑ thoracic kyphosis, ↓ lumbar lordosis

- LoG goes anterior (40mm), Overall ROM ↓,
- Osteoporosis → kyphosis ↑, ↓ Lung volume max
- inspiratory pressure ↓.
- Tight and Lax muscles, balance ↓, wide BOS.
- Balance better in active posture

Gender

- Some experiments showed greater movement of COP in boys than girls.
- This Antero-posterior sway ↓ after (9-10 yrs)
- Some aspects of balance still developing in boys at age of (9-10 yrs).

Pregnancy

- Overall wt gain in anterior region.
- LoG & CoG moves anteriorly.
- Ligamentous laxity → pelvis, SIJ, pubic symphysis, AB
- Lordotic curve ↑ in lumbar and cervical region.
- Anterior pelvic tilt by average of 4°
- Head starts moving back to maintain CoG over BOS
- Foot also sometimes goes for pronation with wide BOS

Occupation

- Every occupatⁿ has few primary postures that are dominant.

→ In recreational activities and sports we tend to develop asymmetrical posture - (Dancing, playing football)

→ Sustained posture in occupation
Over use injury (muscle, tendon, ligament)

→ Back pain, Neck, Shoulder, Nerve disorders are commonly seen in musicians.

→ Each occupation require detailed analysis (Biomechanical)

Industrial therapy.

→ Work site and work place evaluation - H position, light chair position, repetitive movement, Temperature